Ellie (Gaeun) Choi

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Portfolio: ellie-portfolio-psi.vercel.app | github.com/worldstar0722

EDUCATION

M.S. in Finance (MSF) + Graduate Certificate in FinTech Jan 2027 – Exp. Dec 2028

University of Utah, David Eccles School of Business | Salt Lake City, UT

B.S. in Information Systems Exp. Dec 2026

University of Utah, David Eccles School of Business | Salt Lake City, UT

- Minors: Business Analytics and Urban Ecology & Planning
- Relevant coursework: Cloud Computing, Networking & Servers, Data Structures & Java, Visual Analytics, Blockchain for Business, Healthy Urban Planning, GIS for Urban Ecologists, Planning in Indigenous Communities

Visual Administration (Visual Media Management) Diploma Mar 2019 – Jan 2022

Seoul Visual Media High School | Seoul, South Korea

- Specialized curriculum: media production, accounting, marketing, advertising, and visual content creation.

TECHNICAL SKILLS

Data & Analytics: Python (pandas, matplotlib, seaborn, scikit-learn, nltk), SQL/MySQL, Tableau, Power BI, Excel modeling

NLP & Machine Learning: FinBERT, VADER, Random Forest, LightGBM, Gradient Boosting, LLM evaluation (Gemma/Ollama), Yahoo Finance API

Spatial & Urban Analytics: ArcGIS Pro, QGIS, Healthy Places Index (HPI), buffer analysis, Select by Location, spatial scoring, field data collection, land use analysis, urban transect methodology

Programming & Tools: HTML, CSS, Java (data structures), GitHub, Google Colab, LucidChart, Salesforce, Notion, Slack, Microsoft Teams, Google Workspace

Design: Adobe Creative Suite (Photoshop, Illustrator, Premiere Pro, Lightroom, InDesign, Audition, Express), Figma, Canva

RESEARCH & PRESENTATIONS

Finance Chatbot Evaluation: Prompt Trade-offs in Financial Services Fall 2026 – Present

Undergraduate Researcher (with Anne Lee, GaHyun Choi) | Faculty Mentor: Matt Pecsok | Dept. of Operations & Information Systems, David Eccles School of Business | University of Utah

- Building an agent-based evaluation pipeline in which a simulated customer agent converses with a finance-company chatbot and an evaluator agent scores each conversation across multiple metrics.
- Research question: when the chatbot prompt is improved, do all metrics rise together, or do trade-offs emerge?
- Running Gemma 3 (Ollama) in Google Colab with versioned prompts, structured conversation logs, and regression checks in a shared GitHub repository; target: URS poster.

A Multidimensional Framework for Winning Super Bowl Advertising 2026

First Author (with Cienna Kim, Doahn Lee, Hangu Lee) | Faculty Advisor: Chong Oh | Dept. of Operations & Information Systems, David Eccles School of Business; Dept. of Psychology, College of Social & Behavioral Science; Dept. of City & Metropolitan Planning, College of Architecture + Planning | University of Utah

- Analyzed 64,230 tweets from 59 Super Bowl LX brands across in-game and 24-hour post-game windows; cleaned ~30,000 irrelevant tweets.
- Engineered 6 multidimensional metrics — Viral Index, Positive Ratio, Engagement Efficiency, ROI Efficiency ($8M cost baseline), Sustained Ratio, Net Sentiment — normalized into a composite Winning Score Model.
- Proposed a new operationalization of advertising "success": strongest integrated strategic return across all dimensions, not highest mentions or most liked.
- Key findings: Toyota ranked most balanced across all 5 dimensions; $R^2 < 0.05$ between Ad Meter rankings and Winning Score, showing viewer preference and digital engagement capture different outcomes.
- Strategic implication: sustained momentum signals long-term brand imprinting; campaigns should incorporate post-event amplification strategies.

Predicting Future Stock Price Movements Using Historical Market Data and Investor Sentiment 2026

Co-Author (with Maryam Ahmad, Yerke Nokenova, Julia Park) | Faculty Mentors: Emre Ertan, Chong Oh | Dept. of Operations & Information Systems, David Eccles School of Business | University of Utah

- Tested whether investor sentiment adds predictive signal by combining S&P 500 price data (Yahoo Finance API) with FinBERT sentiment from financial news headlines (Google News RSS).
- Trained and compared LightGBM, Random Forest, and Gradient Boosting models with vs. without sentiment features; evaluated with precision, recall, F1, accuracy, and confusion matrices.
- Key findings: models reached ~50% accuracy on next-day direction, highlighting how hard short-horizon prediction is and the need for stronger out-of-sample validation.
- Limitation: headline-level sentiment only; future work may leverage LLMs for full article-level feature extraction.

WORK EXPERIENCE

Teaching Assistant — IS 3060 (Information Systems) Aug 2025 – Present

David Eccles School of Business, University of Utah | Salt Lake City, UT

- Support 160+ students per semester through labs, grading, and 1-on-1 guidance; assess Tableau, Thunkable, and Google Cloud lab work.
- Selected as TA for the 2025 Adobe Social Impact Challenge, advising teams and providing structured feedback on business and design strategy.

Student Finance Operations Assistant 2026 – Present

College of Architecture & Planning, University of Utah | Salt Lake City, UT

- Process and reconcile student finance transactions and operational records, maintaining data integrity across budgeting and reporting workflows.
- Apply accounting fundamentals to flag and resolve discrepancies, supporting accurate financial reporting for the college.
- Build standardized Excel benchmarking workbooks with formula-driven summaries and validate formulas for internal academic-planning datasets.

IS Department Ambassador Jun 2025 – Present

David Eccles School of Business, University of Utah | Salt Lake City, UT

- Serve as liaison between IS students and faculty, supporting outreach events and program recruitment efforts.
- Produce promotional content and collaborate with student clubs to expand IS program visibility across campus.

Marketing Director, International Students in Business (ISB) Club 2025 – 2026

David Eccles School of Business, University of Utah | Salt Lake City, UT

- Led marketing for campus-wide events promoting the international business community and cross-cultural engagement.
- Created marketing content and member communications; supported international students in academic adaptation.

Lead Peer Advisor Feb 2024 – Dec 2024

University of Utah Asia Campus | Incheon, South Korea

- Coordinated peer advising activities, mentored junior advisors, and facilitated workshops on course selection and career planning.
- Provided academic guidance and mentorship to 30+ business students per semester.

Computer Lab Assistant Aug 2023 – Jun 2024

University of Utah Asia Campus | Incheon, South Korea

- Provided technical assistance including hardware/software troubleshooting, networking support, and printing/scanning help for students and faculty.
- Maintained lab operations and escalated issues to IT department as needed.

Vice President, Accounting Club Jun 2024 – Dec 2024

University of Utah Asia Campus | Incheon, South Korea

- Led club operations and organized professional development events focused on accounting and finance careers.
- Facilitated open tutoring sessions to support students struggling with accounting coursework.

Vice President, TEK (IS) Club Jul 2023 – Dec 2024

University of Utah Asia Campus | Incheon, South Korea

- Planned and led club meetings and workshops on IS topics: data management, database design, and software development.
- Collaborated with faculty and student organizations to promote information systems education and research.

Internship — Travel & Cultural Programming Jul 2023

Kim's M&T | Seoul, South Korea

- Planned Korea cultural travel itineraries for foreign clients.
- Served as assistant staff for World Scout Jamboree 2023 and Manchester City Asia Tour.

Triple University Supporters — Content Creator Jul 2023

Interpark Triple Corp. | Seoul, South Korea

- Created multimedia travel content — video production, blog writing, and original music composition — for promotional campaigns.

Barista & Shift Manager Jan 2022 – Aug 2022

Nolsoop | Gwangyang, South Korea

- Managed daily café operations: customer service, order taking, payment processing, inventory, and cleanliness standards.

ACADEMIC PROJECTS

Uber Ride-Hailing Database Prototype Summer 2026

IS 4420: Database Fundamentals | Team: Ellie Choi, Richard Conti, Julia Park | University of Utah

- Designed conceptual, logical, and physical ERDs (LucidChart) for a 17-entity MySQL database; exported DDL and documented business rules and conflicting requirements.
- Team delivered DDL/DML scripts, synthetic sample data, and 10 analytical SQL queries on trips, payments, ratings, and driver performance.

Athlete Housing Search & Residence Approval System Summer 2026

IS 4430: Systems Analysis & Design | Team of 4 | University of Utah

- Co-developed requirements, data-flow diagrams, ERD, and trigger/domain catalogs for an athlete housing search and residence-approval system.
- Team delivered an interactive multi-role UI prototype (athlete, reviewer, admin) with budget comparison and approval workflow.

Where Should You Invest in Salt Lake City? — GIS Investment Screening Analysis Summer 2026

CMP 4450: GIS for Urban Ecologists | University of Utah

- Scored all 2,711 SLC residential zones on a 0–3 composite walkability index combining access to schools (1-mile buffer), parks (0.5-mile), and TRAX light rail stations (0.5-mile) using ArcGIS Pro and five public UGRC datasets.
- Applied Select by Location (Intersect) + Calculate Field to flag each zone per amenity; exported results to Excel for summary statistics.
- Key findings: 713 zones (26.3%) achieved full access (score 3); transit is the bottleneck — only 742 zones reach a TRAX station vs. 2,678 reaching a school and 2,652 reaching a park.
- Score-3 zones trace the TRAX corridor (Downtown, Central City, Liberty Wells, Ballpark, Sugar House); west-side gap is a transit gap, not an amenity gap.
- Limitations: straight-line buffers rather than street-network distance; no home-price data, so the score is a screening proxy, not a value estimate.

SLC Civic Center Healthy Urban Planning Analysis (Green Loop) Spring 2026

CMP 4060: Healthy Urban Planning | with Hannah Dudley | University of Utah

- Assessed the Civic Center within Salt Lake City's Green Loop context using site observations, HPI comparisons, and pedestrian counts to evaluate equity, green-space access, and chronic-disease risk.
- Proposed traffic calming, active-use green space programming, and anti-gentrification policy interventions to improve pedestrian safety outcomes.

Urban Transect Field Study — Salt Lake City Spring 2026

CMP 3200 | University of Utah

- Conducted field-based urban transect across SLC neighborhoods using QGIS and field photography, documenting pedestrian infrastructure, bike lane access, green space quality, and land use patterns.
- Synthesized observations into visual maps and written analysis examining the tension between vehicular infrastructure and active transportation design.

Musqueam First Nation Community Planning Analysis Summer 2025

CMP 6960: Planning in Indigenous Communities | University of Utah

- Analyzed the xʷməθkʷəy̓əm (Musqueam) Nation's Comprehensive Community Plan across land use, transit, housing equity, energy sovereignty, and environmental stewardship.
- Authored sections on Indigenous foodscapes, energy sovereignty, and landscape governance; co-presented findings to class.

Salesforce-Eccles Ideathon — Athletics Recruiting CRM Sep 2025

Team: Ellie Choi, Doahn Lee, Eubon Shin | David Eccles School of Business, University of Utah

- Co-developed a Salesforce CRM prototype to manage athlete prospects, communications, campus visits, offers, and evaluations through a 5-stage recruiting pipeline.
- Delivered a Kanban pipeline, automated follow-up reminders, role-based dashboards, and a user guidebook.

Lichen Bio-indicator Environmental Field Study Spring 2024

CMP 2010 | University of Utah Asia Campus | Incheon, South Korea

- Collected and analyzed lichen samples as environmental bio-indicators, assessing pollution levels across roadside and park sites in Incheon, Korea; documented findings with field photography and data tables.

CREATIVE WORK — MUSIC DIRECTION

Short Film "HELLPER" — Music Director Sep 2023 – Jul 2024

Bucheon International Fantastic Film Festival (BIFAN) — Official Selection

- Composed and curated original soundtrack to complement the film's narrative and visual tone; film officially screened at BIFAN 2024.

Short Film "구제 (Salvation)" — Music Director Mar 2023 – Sep 2023

Collaborative production with Film and Media Arts students, University of Utah Asia Campus

- Collaborated on writing, casting, and production; composed and directed the film's musical score.

Short Film "먹이 (Feed)" — Music Director Winter 2022 – Spring 2023

Collaborative production with Film and Media Arts students, University of Utah Asia Campus

- Collaborated on writing, casting, and production; composed and directed the film's musical score.

AWARDS & SCHOLARSHIPS

- 1st Place, UAC Adobe Social Impact Challenge — designed webpage layouts and marketing assets using Adobe Creative Suite for a company's branding and social impact strategy (Spring 2023).

- New Horizon Scholarship — $2,000/year (Spring 2023 – Present).
- IS Game Day Analytics Challenge — Participant & Research Poster Presenter (Feb 2026).
- Salesforce-Eccles Ideathon — Participant (Sep 2025).
- Selected for Promotional Video, University of Utah Asia Campus Business Department (April 2024).
- UAC Big Red Event Volunteer (May 2023).

LANGUAGES

Korean (Native) **English** (Professional Working Proficiency)